Bayesian Filter and Particle Filter

Qianshu ZHANG

City University of Hong Kong

Sep 27th, 2023

Introduction

- State–space models are common in statistics, engineering, economics and finance. There are two closely related statistical inference problems in these models, i.e., state filtering and parameter estimation.

Bayesian learning in state–space models

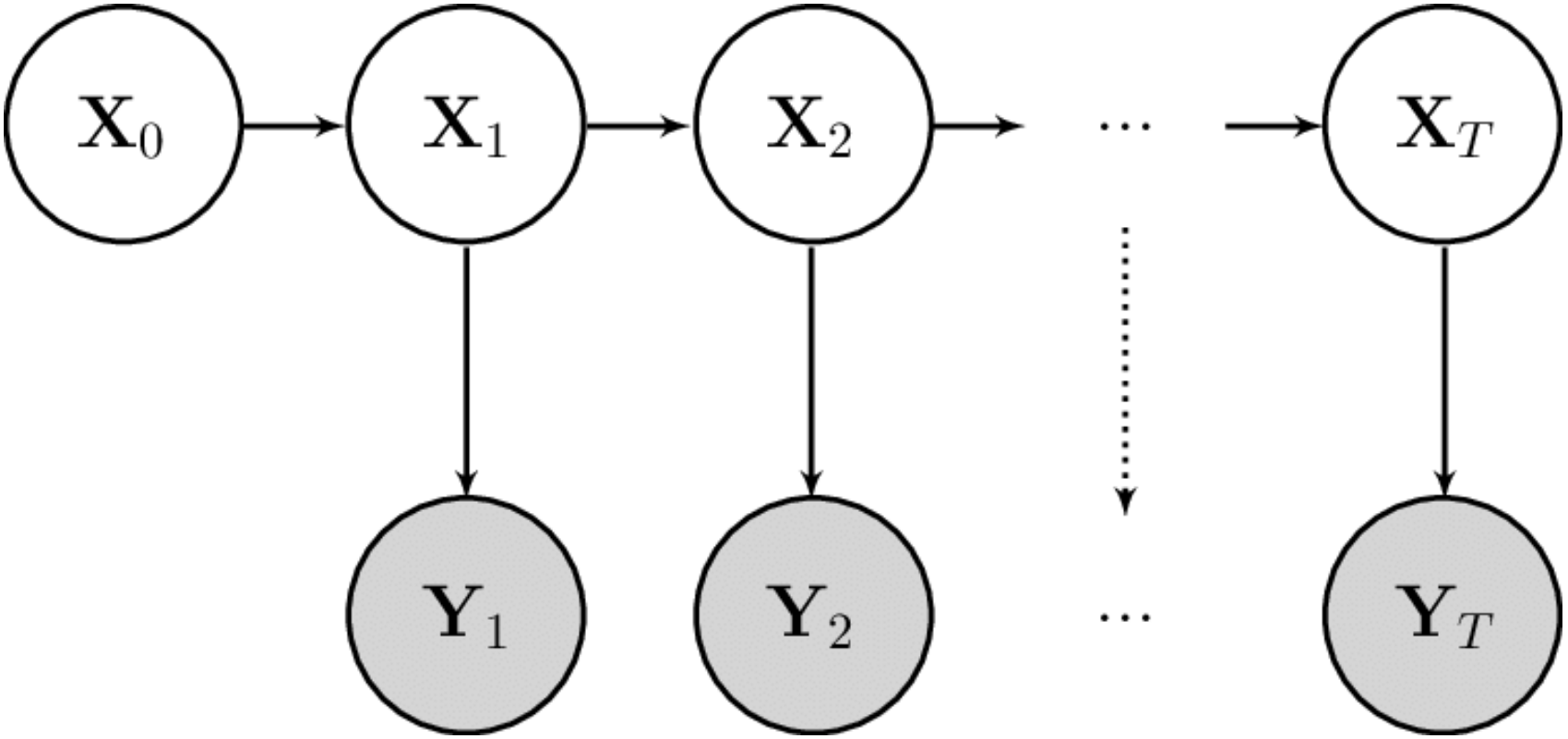
A state–space model can be written in the following general form

$$y_t = H(x_t, w_t, \theta) \tag{1}$$

$$x_t = F(x_{t-1}, v_t, \theta) \tag{2}$$

where the observation y_t is assumed to be conditionally independent given the state x_t with the distribution $p(y_t|x_t, \theta)$, the state x_t is modeled as a Markov process that lives on a space χ with the initial distribution $p(x_0|\theta)$and the transition law $p(x_t|x_{t-1}, \theta)$, w_t and v_t are the mutually independent observation noise and state noise and $\theta \in \Theta$ is a set of the static parameters.

State Space Model



Bayesian learning in state–space models

The task of state filtering and parameter learning is to find the joint posterior distribution of states and parameters based on past and current observations,

$$p(x_t, \theta | y_{1:t}) = p(x_t | y_{1:t}, \theta) p(\theta | y_{1:t}) \tag{3}$$

where $p(x_t | y_{1:t}, \theta)$ solves the state filtering problem, and $p(\theta | y_{1:t})$ addresses the parameter inference issue.

Bayesian filter

- Let Y_t denote $y_{1:t}$, for State filtering, our target is to find the density $p(x_t|Y_t)$ by recursion given $p(x_{t-1}|Y_{t-1})$.
- First find $p(x_t|Y_{t-1})$:

$$p(x_t|Y_{t-1}) = \int p[(x_t, x_{t-1}|Y_{t-1})]dx_{t-1}$$
$$= \int p[x_t|(x_{t-1}, Y_{t-1})]p(x_{t-1}|Y_{t-1})dx_{t-1}$$

- Since x_t is determined only by x_{t-1}, $p[x_t|(x_{t-1}, Y_{t-1})] = p(x_t|x_{t-1})$

$$\Rightarrow p(x_t|Y_{t-1}) = \int p(x_t|x_{t-1})p(x_{t-1}|Y_{t-1})dx_{t-1}$$

Bayesian filter

- Then consider $p(x_t|Y_t)$:

$$p(x_t|Y_t) = \frac{p(Y_t|x_t)}{p(Y_t)}p(x_t) = \frac{p(y_t, Y_{t-1}|x_t)}{p(y_t, Y_{t-1})}\frac{p(x_t|Y_{t-1})p(Y_{t-1})}{p(Y_{t-1}|x_t)}$$
$$= \frac{p(x_t, y_t, Y_{t-1})}{p(x_t)p(y_t, Y_{t-1})}\frac{p(x_t, Y_{t-1})p(Y_{t-1})}{p(Y_{t-1})p(Y_{t-1}|x_t)}$$

where the second part use the identity of conditional probability formula:

$$p(x_t)p(Y_{t-1}|x_t) = p(x_t, Y_{t-1}) = p(x_t|Y_{t-1})p(Y_{t-1})$$

Bayesian filter

- Multiply the equation $p(x_t, y_t)$ up and down simultaneously
$$p(x_t|Y_t) = \frac{p(x_t, y_t, Y_{t-1})}{p(x_t)p(y_t, Y_{t-1})} \frac{p(x_t, Y_{t-1})p(Y_{t-1})}{p(Y_{t-1})p(Y_{t-1}|x_t)} \frac{p(x_t, y_t)}{p(x_t, y_t)}$$
- Simplify using conditional probability formula several times

$$p(x_t|Y_t) = \frac{p(Y_{t-1}|x_t, y_t)p(y_t|x_t)p(x_t|Y_{t-1})}{p(y_t|Y_{t-1})p(Y_{t-1}|x_t)}$$
- notice that y_t is function of x_t, so $p(Y_{t-1}|x_t, y_t) = p(Y_{t-1}|x_t)$,then

$$p(x_t|Y_t) = \frac{p(y_t|x_t)p(x_t|Y_{t-1})}{p(y_t|Y_{t-1})}$$

Bayesian filter

- Last step, find $p(y_t|Y_{t-1})$

$$\begin{aligned} p(y_t|Y_{t-1}) &= \int p(y_t, x_t|Y_{t-1})dx_t \\ &= \int p(y_t|x_t, Y_{t-1})p(x_t|Y_{t-1})dx_t \\ &= \int p(y_t|x_t)p(x_t|Y_{t-1})dx_t \end{aligned}$$

- then

$$p(x_t|Y_t) = \frac{p(y_t|x_t)\int p(x_t|x_{t-1})p(x_{t-1}|Y_{t-1})dx_{t-1}}{\int p(y_t|x_t)p(x_t|Y_{t-1})dx_t}$$

Bayesian filter

- It seems that all the density is known, but the analytical solution is only obtainable under some condition.
- If functions $H(.)$ and $F(.)$ are linear in x_t and w_t and if Gaussian distributions are assumed for x_0, w_t and v_t , the well-known Kalman filter can be applied.
- Don't worry about the functions above. Particle filter is a simulation-based solution to Bayesian filter.

Monte Carlo

- Recursion: if we can approximate $p(x_{t-1}|Y_{t-1})$, we can solve $p(x_t|Y_t)$ by recursion.
- Monte Carlo approximation is a good way to find the $p(x_t|Y_t)$ if we can get samples from it.
- Unfortunately, it is usually impossible to sample efficiently from the posterior distribution at any time t, being multivariate, non-standard, and only known up to a proportionality constant.

Importance Sampling

- We can smaple from easily sampled distribution and assign sample weight, then weighted sample can summarize the target distribution.

$$E[f] = \int f(z)p(z)dz$$

$$E[f] = \int f(z)\frac{p(z)}{q(z)}q(z)dz$$

- Sequential importance smapling: if we can approximate $p(x_{t-1}|Y_{t-1})$ by samples of weight $w_{t-1} = \frac{p(x_{1:t-1}|y_{1:t-1})}{q(x_{1:t-1}|y_{1:t-1})}$, and find the relationship between w_t and w_{t-1}, we can solve recursively.

Sequential Importance Sampling

$$
\begin{aligned}
w_t &= \frac{p(x_{1:t}|y_{1:t})}{q(x_{1:t}|y_{1:t})} \\
p(x_{1:t}|y_{1:t}) &= \frac{p(x_{1:t}, y_{1:t})}{p(y_{1:t})} = \frac{1}{C} p(x_{1:t}, y_{1:t}) \\
&= \frac{1}{C} p(y_t|x_{1:t}, y_{1:t-1}) p(x_{1:t}, y_{1:t-1}) \\
&= \frac{1}{C} p(y_t|x_{1:t}, y_{1:t-1}) p(x_t|x_{1:t-1}, y_{1:t-1}) p(x_{1:t-1}, y_{1:t-1}) \\
&= \frac{1}{C} p(y_t|x_{1:t}, y_{1:t-1}) p(x_t|x_{1:t-1}, y_{1:t-1}) \\
&\quad p(x_{1:t-1}|y_{1:t-1}) p(y_{1:t-1}) \\
&= \frac{D}{C} p(y_t|x_t) p(x_t|x_{t-1}) p(x_{1:t-1}|y_{1:t-1})
\end{aligned}
$$

Sequential Importance Sampling

$$
\begin{aligned}
q(x_{1:t}|y_{1:t}) &= q(x_t|x_{1:t-1}, y_{1:t})q(x_{1:t-1}|y_{1:t}) \\
&= q(x_t|x_{1:t-1}, y_{1:t})q(x_{1:t-1}|y_{1:t-1})
\end{aligned}
$$

then

$$
\begin{aligned}
w_t &= \frac{p(x_{1:t}|y_{1:t})}{q(x_{1:t}|y_{1:t})} \\
&\propto \frac{p(y_t|x_t)p(x_t|x_{t-1})p(x_{1:t-1}|y_{1:t-1})}{q(x_t|x_{1:t-1}, y_{1:t})q(x_{1:t-1}|y_{1:t-1})} \\
&\propto \frac{p(y_t|x_t)p(x_t|x_{t-1})}{q(x_t|x_{1:t-1}, y_{1:t})} w_{t-1}
\end{aligned}
$$

Sequential Importance Sampling

- If we let proposal distribution equals to $p(x_t|x_{t-1})$ then

  $w_t \propto p(y_t|x_t)w_{t-1}$
- We can solve recursively.
- Up to now, we only consider state filtering, and take parameter as known.

Solve State filtering and parameter learning jointly

- Chopin, Nicolas, Pierre E. Jacob, and Omiros Papaspiliopoulos. SMC^2: an efficient algorithm for sequential analysis of state space models." Journal of the Royal Statistical Society Series B: Statistical Methodology 75, no. 3 (2013): 397-426.
- This paper gives the justification of sample state and parameter jointly.

Particle filter

- Suppose we randomly generate N state samples and N*K parameter(suppose paramter set θ contains K parameters) samples based on the initial probability density. At each time point, we can use the state transition equation $p(x_t|x_{t-1}, \theta)$ to obtain the state particles in the next state.
- At time t, after obtaining the observation value y, we can calculate the likelihood $p(y_t|x_t, \theta)$ of each particle and update particle weight.

Problem of Particle Filter

- Effective sample size(ESS):

$$ESS = \frac{1}{\sum_{i=1}^{N}(w_i)^2}$$

- Example:

  All weights $= 1/N \rightarrow$ ESS $= N$

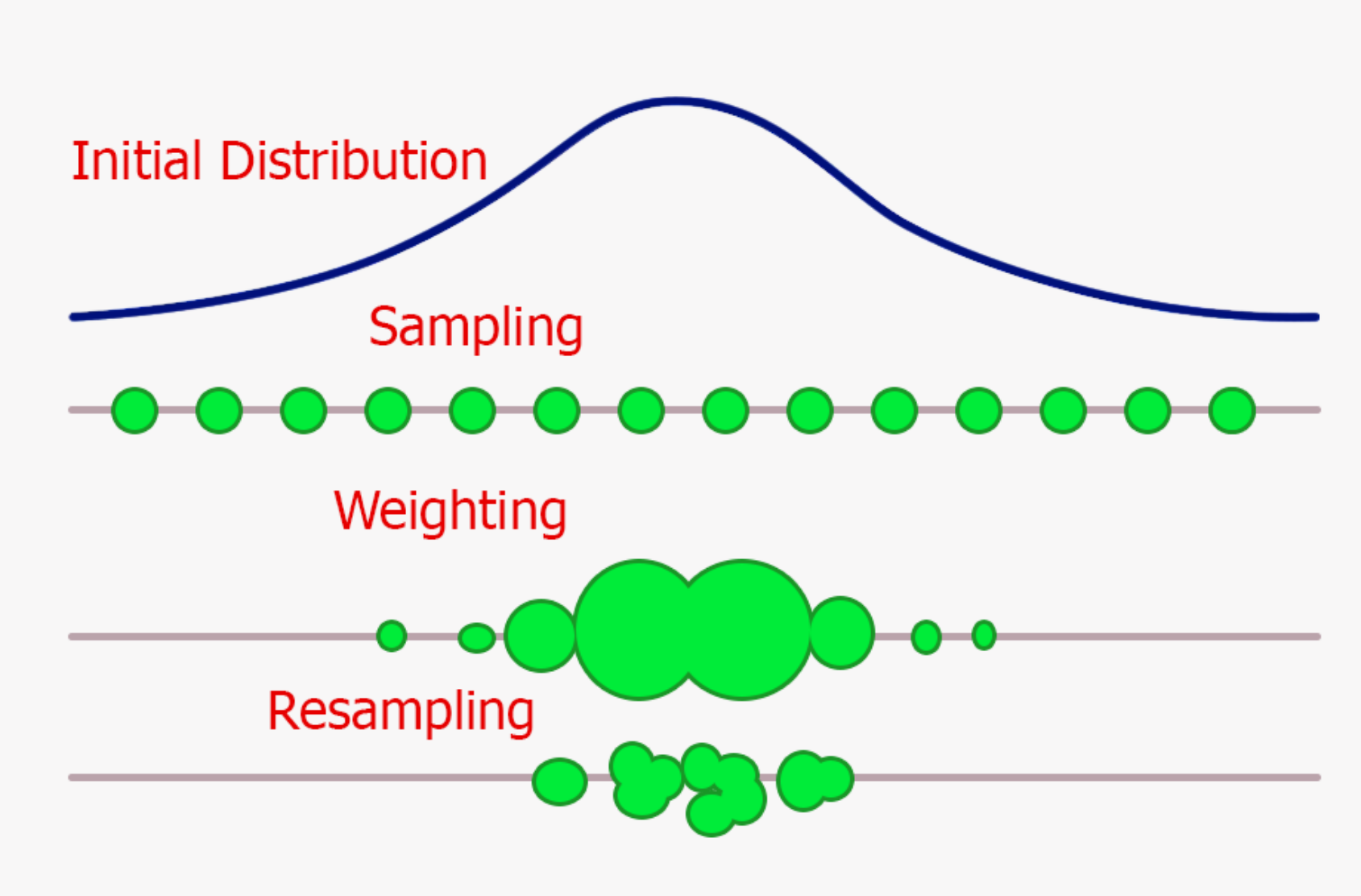
  All weights $= 0$, except for one weight $= 1 \rightarrow$ ESS $= 1$

- When ESS is low, many particles are useless with a low weight. We should perform resample.

Resample and Move

- When ESS $<$ some threshold, resample the particles proportional to w_i and provide an equally-weighted sample
- But this will lead to sample deterioration. In the end there will only be one particle with the highest weight.
- Move step:
  Move each particle through a Markov kernel with a proposal stationary distribution while the number of unique particles is below some other threshold.
- We can imagine adding a small disturbance to each particle to add different particles without changing the overall particle distribution.

Resample and Move



Simulation settings

- model:

$$r_t = \mu + \epsilon_t, \epsilon_t \sim N(0, h_t),$$
$$h_t = \omega + \alpha\epsilon_{t-1}^2 + \beta h_{t-1}$$

- here dynamics of h_t follows a GARCH. r_t is observable, h_t is the hidden state. Parameter set includes $\{\mu, \omega, \alpha, \beta\} = \{0.10/252, 0.025/252, 0.02, 0.95\}$, initial value of h equals to 0.025/252

Simulation results

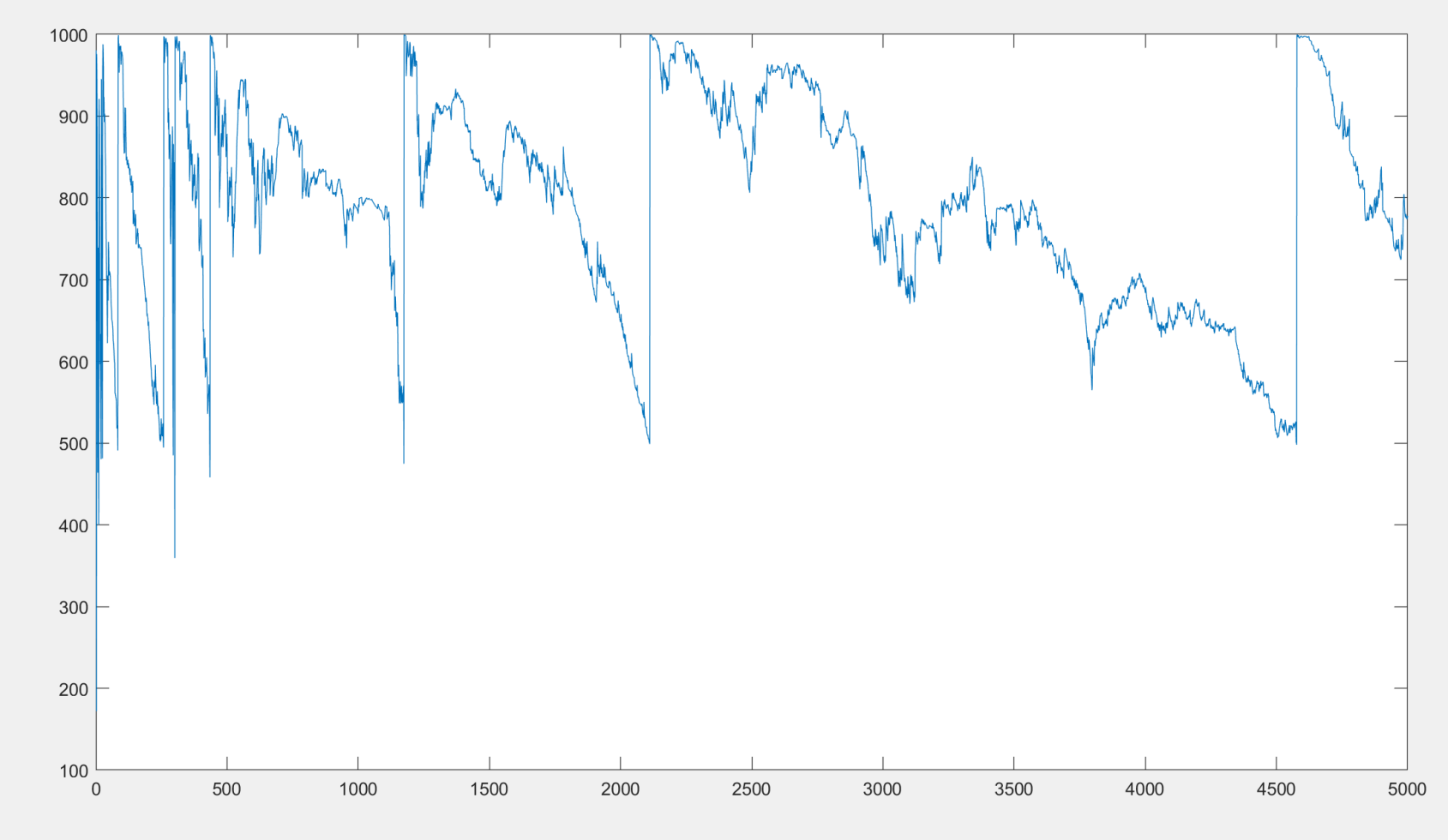


FIGURE: Effective Sample Size

Simulation results

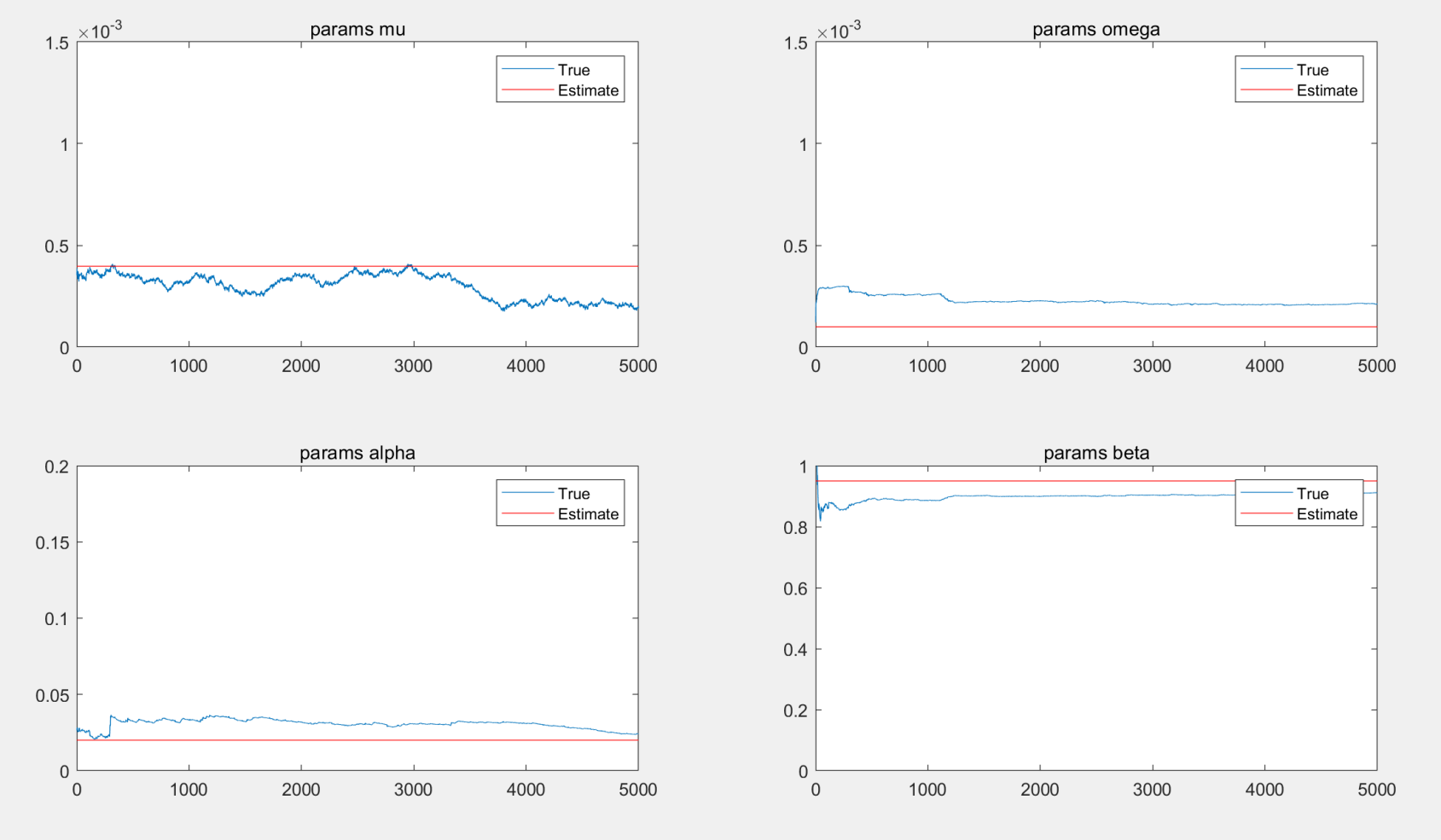


FIGURE: Fix parameter estimation

Simulation results

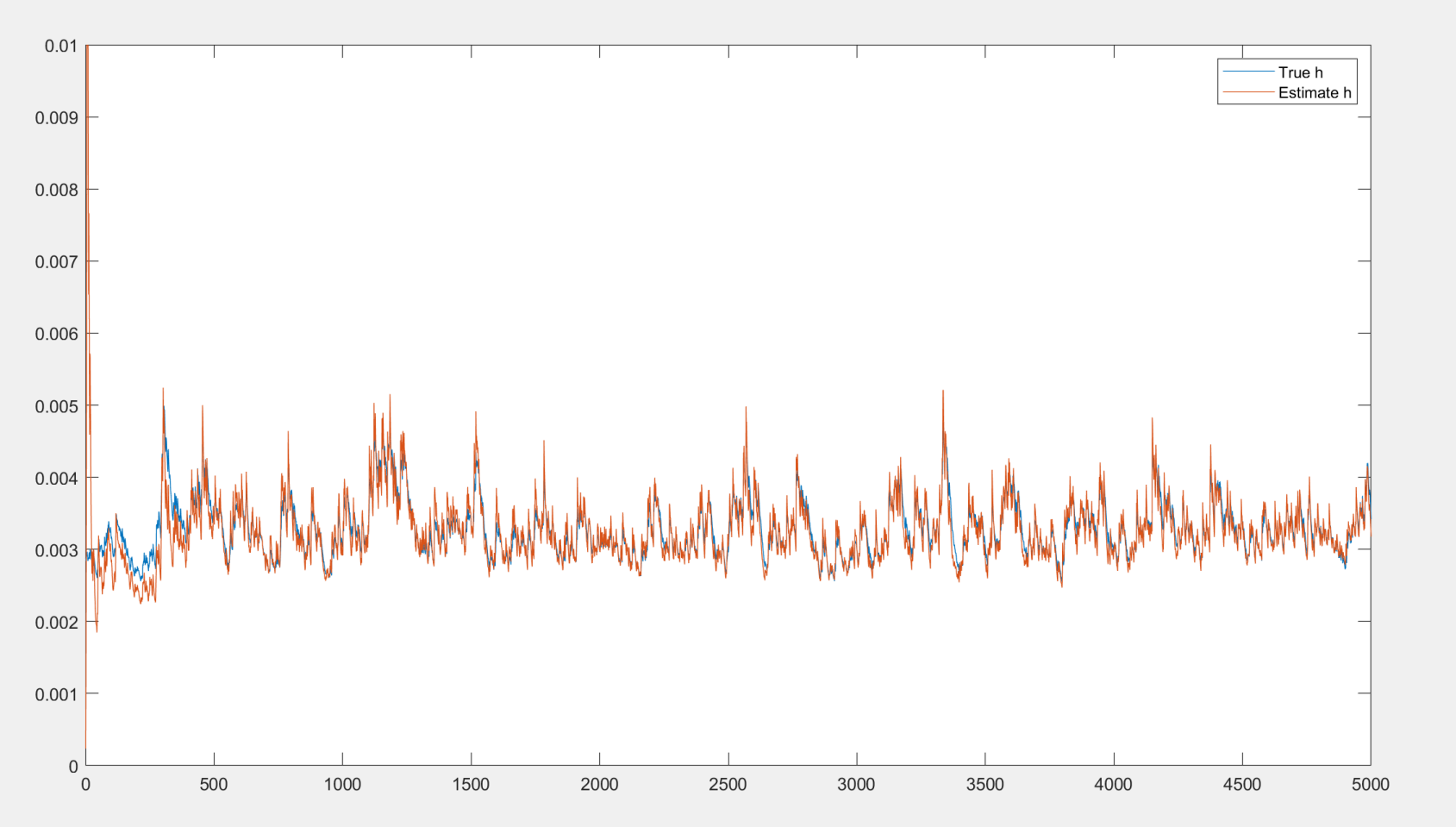


FIGURE: State estimation

Thanks!